Kaplan turbine (in-stream) — manufacturing drawing set

Generated from turbines/freecad/kaplan.FCStd · every view, section and dimension measured from the model

DRAWING LIST

No.	Title	Rev.
TL-KAP-000	Cover, drawing list, general notes	A
TL-KAP-001	General assembly and bill of materials	A
TL-KAP-002	Runner assembly	A
TL-KAP-003	Runner blade	A
TL-KAP-004	Blade trunnion	A
TL-KAP-005	Runner hub	A
TL-KAP-006	Bulb nacelle	A
TL-KAP-007	Duct (shroud)	A
TL-KAP-008	Guide vane	A
TL-KAP-009	Support pylon and gravity base	A
TL-KAP-010	Manufacturing process plan — runner	A
TL-KAP-011	Manufacturing process plan — structure	A
TL-KAP-012	Assembly sequence, inspection and test plan	A

GENERAL NOTES

1. Dimensions in millimetres. Model coordinates: X along the runner axis (flow +X), Z up; the runner axis is at Y = 0, Z = -1800.

2. General tolerances ISO 2768-mK for machined parts; ISO 13920-BF for weldments; ISO 8062-3 DCTG 10 for cast surfaces not machined.

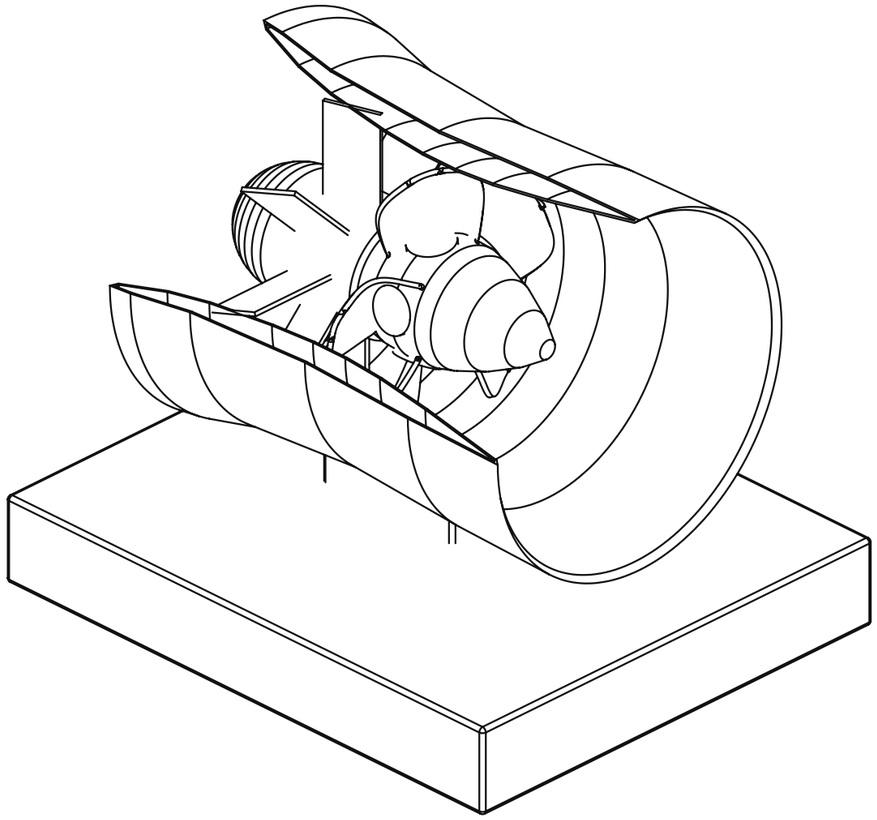
3. Welding to EN ISO 3834-2; procedures to EN ISO 15614-1; welders to EN ISO 9606-1; quality level B to EN ISO 5817 for load-bearing welds.

4. Corrosion protection for permanent immersion in fresh water: blast Sa 2½ (ISO 8501-1), epoxy system for ISO 12944 class Im1, plus sacrificial anodes.

5. Masses are calculated from the model volumes, or for parts the model shows solid (hub, bulb, duct, pylon) from its area and the wall thickness stated in the bill of materials.

6. The model's duct is a 270° cutaway for display; it is manufactured as a full 360° ring.

7. This is a teaching model: materials, processes and tolerances are typical choices for a small hydrokinetic Kaplan unit, for learning, and are not a design. Not for construction.



Isometric view of the unit (not to scale)

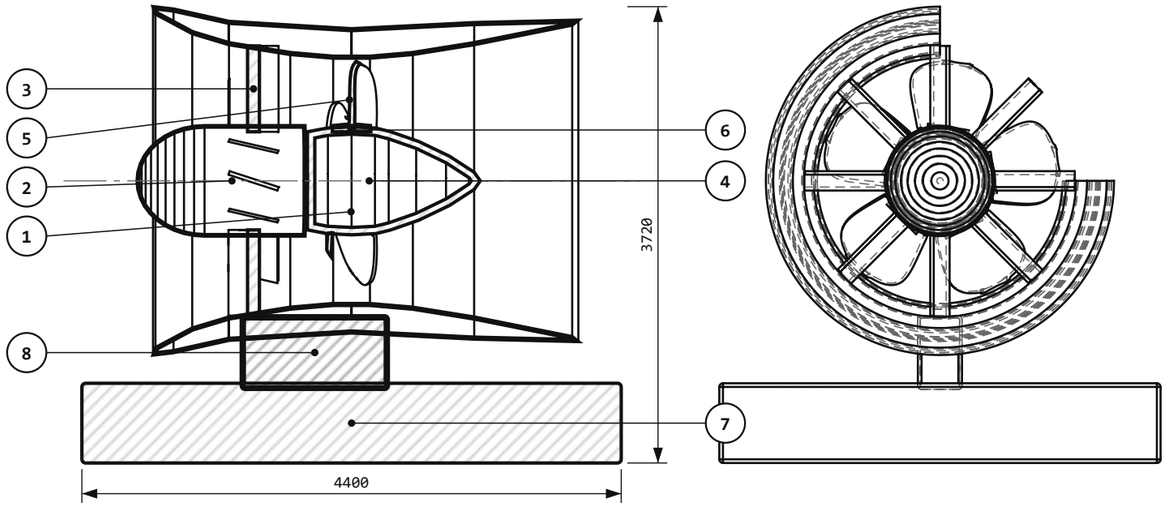
Revision A · 2026-11-06 · first issue

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

KAPLAN TURBINE — MANUFACTURING PLANS

In-stream Kaplan unit, Turbine Lab FreeCAD model · drawing set and process plan

Drawing no. **TL-KAP-000**	Sheet **1 / 13**	Scale (A3) **—**	Rev. **A**
Material **—**	Mass **—**	Units **mm**	Date **2026-11-06**
General tol. **ISO 2768-mK**	Welds **ISO 13920-BF**	Projection	Drawn **Turbine Lab**

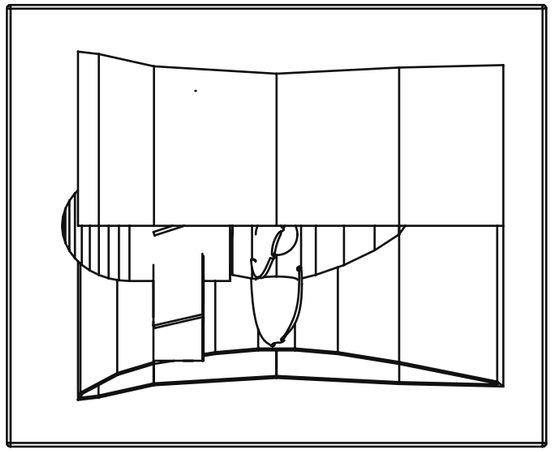


SECTION A–A

through the runner axis · scale 1:50

VIEW B

from upstream · scale 1:50



PLAN

scale 1:50

BILL OF MATERIALS

Item	Part	Qty	Material	Process	Unit	Total
1	Duct (shroud)	1	S355J2+N plate, EN 10025-2	Rolled plate segments, welded, stiffened	7.65 t	7.65 t
2	Bulb nacelle	1	S355J2+N plate, EN 10025-2	Formed plates, welded, watertight	488 kg	488 kg
3	Guide vane	8	S355J2+N plate, EN 10025-2	Plate, cut and formed	69 kg	554 kg
4	Runner hub	1	G20Mn5+QT cast steel, EN 10293	Cast, machined	1.48 t	1.48 t
5	Runner blade	5	CA6NM (13Cr-4Ni), ASTM A743	Cast, 5-axis machined	105 kg	524 kg
6	Blade trunnion	5	42CrMo4+QT, EN 10083-3	Forged, turned	36 kg	178 kg
7	Gravity base	1	Concrete C35/45, B500B rebar	Precast, reinforced	24.69 t	24.69 t
8	Support pylon	1	S355J2+N plate, EN 10025-2	Welded box section	391 kg	391 kg

Total mass 35.96 t

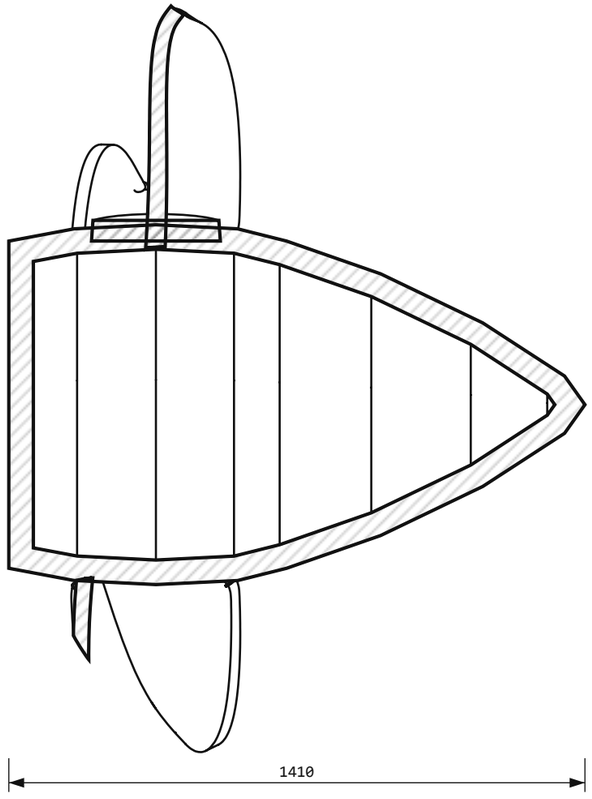
1. Unit masses from the model volume, except: duct (shroud) — 16 mm skin + 15 % stiffeners, full 360°; bulb nacelle — 12 mm shell + 20 % frames; runner hub — 60 mm cast wall; support pylon — 20 mm plate box.
2. Section A–A cuts the unit on the vertical plane through the runner axis; the far half is shown, cut faces hatched.
3. Flow from left to right (+X). Blades shown at the model's blade angle.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

GENERAL ASSEMBLY

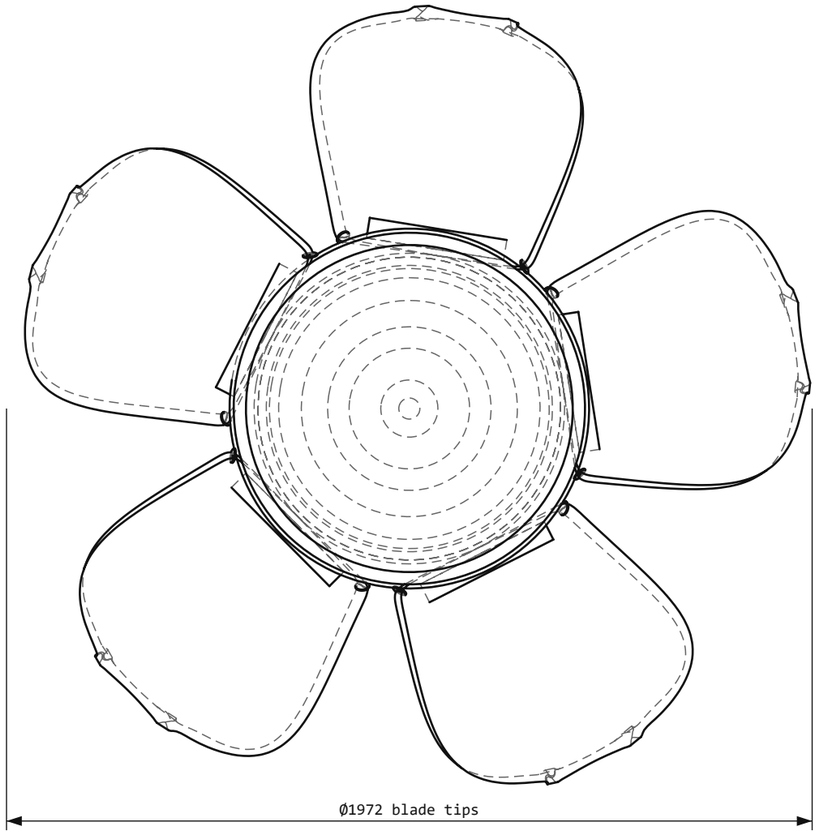
Section through the runner axis, end view, plan; bill of materials

Drawing no. **TL-KAP-001**	Sheet **2 / 13**	Scale (A3) **—**	Rev. **A**
Material **see BOM**	Mass **35.96 t**	Units **mm**	Date **2026-11-06**
General tol. **ISO 2768-mK**	Welds **ISO 13920-BF**	Projection	Drawn **Turbine Lab**



SECTION C–C

through the runner axis · scale 1:15



VIEW D

from upstream · scale 1:15

RUNNER DATA

Quantity	Value
Number of blades	5
Blade spacing	72° (trunnions at 27°, 99°, 171°, 243°, 315°)
Runner diameter (blade tips)	Ø1972
Hub diameter	Ø880
Hub ratio	0.45
Blade mass (each)	105 kg
Runner mass (hub, blades, trunnions)	2.18 t

REQUIREMENTS

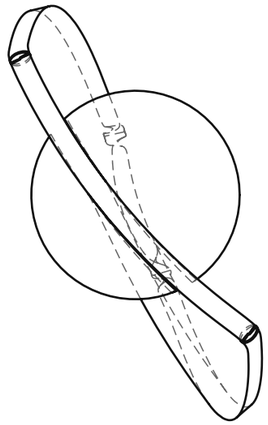
1. Blades fitted to the hub on their trunnions; all five set to the same angle within ±0.1°, checked with the blade-angle gauge at the tip section.
2. Tip clearance to the duct 3 ± 1 mm all round, measured at 8 positions after installation.
3. Balance the assembled runner to ISO 21940-11, grade G6.3, at the maximum service speed; correction by grinding the hub, not the blades.
4. Blade-to-hub and trunnion bolting torques and the pitch mechanism inside the hub to be designed; not shown in this model.
5. Mark each blade and its hub seat with the same number 1–5 before balancing; do not interchange after balancing.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

RUNNER ASSEMBLY

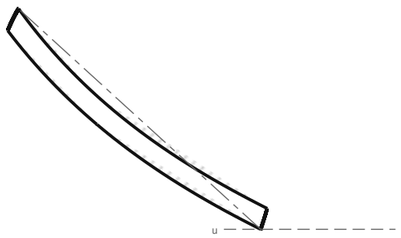
Hub, five blades and trunnions

Drawing no. TL-KAP-002	Sheet 3 / 13	Scale (A3) —	Rev. A
Material see items 4–6	Mass 2.18 t	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab



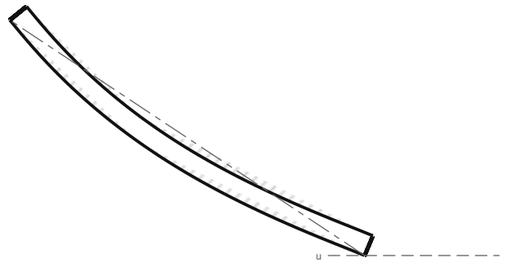
VIEW E

down the pivot axis, from the tip · scale 1:10



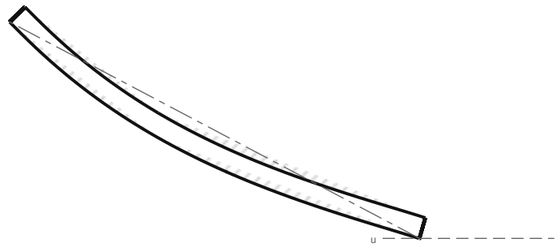
SECTION G — r = 428 · 1:10

chord 534 · β 42.3° · t 36



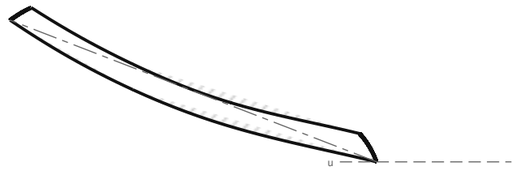
SECTION H — r = 592 · 1:10

chord 695 · β 33.5° · t 37



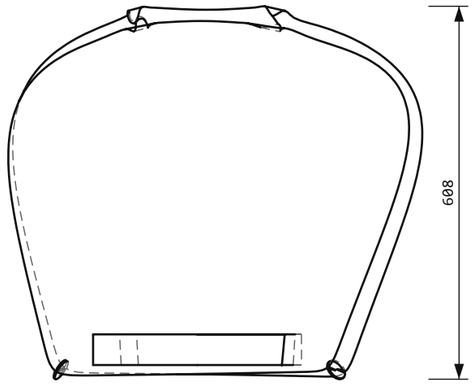
SECTION J — r = 756 · 1:10

chord 756 · β 27.8° · t 37



SECTION K — r = 925 · 1:10

chord 641 · β 21.1° · t 39



VIEW F

along the blade chord direction · scale 1:10

SECTION DATA (developed on cylinders about the runner axis)

Section	Radius r	r / tip	Chord	Blade angle β	Thickness
G	428	0.43	534	42.3°	36
H	592	0.60	695	33.5°	37
J	756	0.77	756	27.8°	37
K	925	0.94	641	21.1°	39

1. Sections are developed (unrolled) cylindrical cuts at radius r about the runner axis (u: circumferential direction, flow upward); β is the chord angle to the circumferential direction. The blade is a plate of uniform thickness, edges rounded.

2. Profile tolerance of blade surfaces ±1.0 mm from the CAD model (step/blade.step); leading and trailing edges checked with templates at G–K.

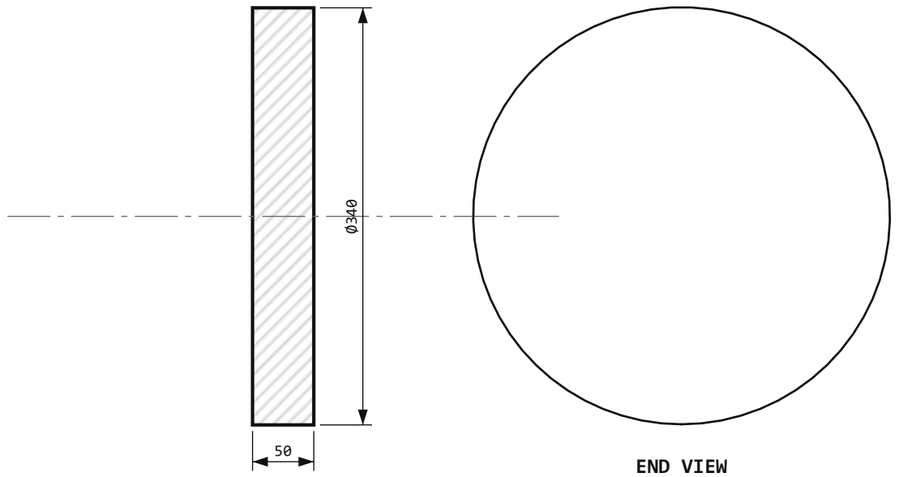
3. Surface finish Ra ≤ 3.2 µm on both faces; edges radiused as modelled. Cast CA6NM, normalised and tempered; 100 % PT (ISO 3452) after machining; UT of the root (ISO 17640 / ASTM A609).

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

RUNNER BLADE

Developed cylindrical sections, plan and front view

Drawing no. TL-KAP-003	Sheet 4 / 13	Scale (A3) 1:10	Rev. A
Material CA6NM (13Cr-4Ni), ASTM A743	Mass 105 kg	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab



SECTION

on the axis · scale 1:5

END VIEW

scale 1:5

PROFILE COORDINATES

Axial position a from the first face; diameters on the trunnion axis.

Pt	Axial a	Ø = 2r
1	50.0	340.0
2	0.0	340.0

1. Forged 42CrMo4+QT (EN 10083-3), 900–1100 MPa; UT of the forging to EN 10228-3 class 3.

2. Turn all over; faces flat within 0.05 mm and parallel within 0.05 mm; seat diameter to h7 against the hub bore (H7), fit to be confirmed in design.

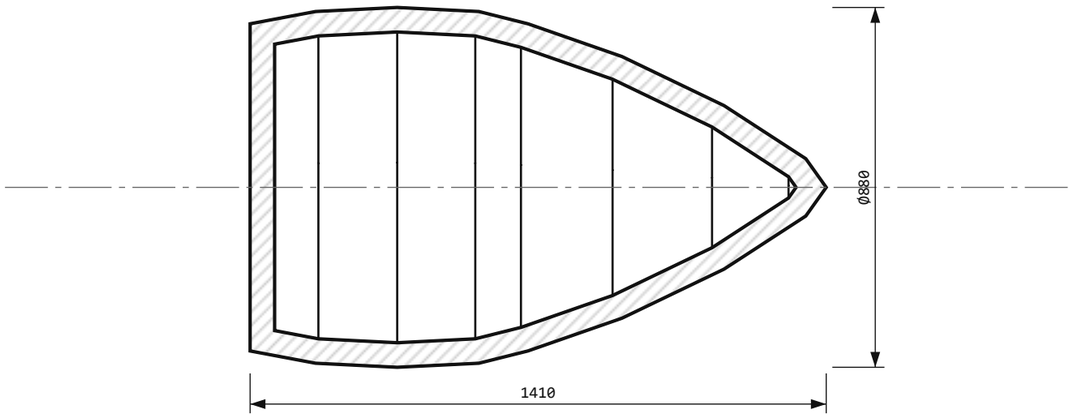
3. Bolting pattern to the blade and the pitch-link pin are not in the model; to be designed. Ra 1.6 µm on seat faces, 3.2 µm elsewhere.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

BLADE TRUNNION

Turned flange carrying the blade on the hub

Drawing no. TL-KAP-004	Sheet 5 / 13	Scale (A3) 1:5	Rev. A
Material 42CrMo4+QT, EN 10083-3	Mass 36 kg	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab



SECTION

on the axis · scale 1:15

PROFILE COORDINATES

Axial position a from the upstream end; outer surface of the model (cast wall 60 mm assumed).

Pt	Axial a	∅ = 2r
1	910.0	640.0
2	680.0	800.0
3	560.0	860.0
4	360.0	880.0
5	160.0	860.0
6	0.0	800.0
7	0.0	0.0
8	1410.0	0.0
9	1360.0	140.0
10	1160.0	400.0

TRUNNION SEATS

Seat	Angle θ	Axial x	Radius
1	99.2°	0.0	420.0
2	171.2°	0.0	420.0
3	243.2°	0.0	420.0
4	315.2°	0.0	420.0
5	27.2°	0.0	420.0

1. Cast G20Mn5+QT (EN 10293); casting tolerance ISO 8062-3 DCTG 10; MT of machined surfaces (ISO 17638), UT of seat zones.

2. Seat bores H7 on radial axes at the angles above, true position Ø0.1 mm to the hub axis; seat faces square to their axis within 0.05 mm.

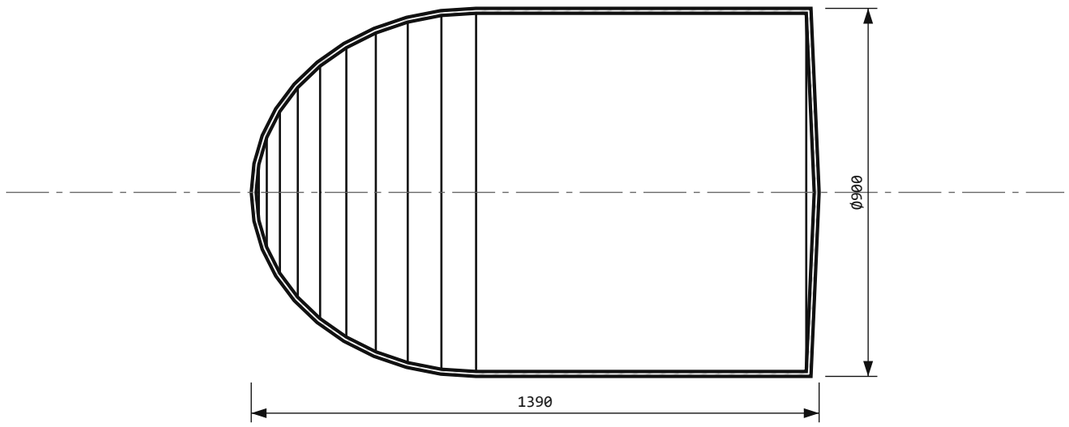
3. Hub interior (pitch mechanism) and the shaft flange are not in the model; to be designed.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

RUNNER HUB

Cast hub with five trunnion seats

Drawing no. TL-KAP-005	Sheet 6 / 13	Scale (A3) 1:15	Rev. A
Material G20Mn5+QT cast steel, EN 10293	Mass 1.48 t	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab



SECTION

on the axis · scale 1:15

PROFILE COORDINATES

Axial position a from the upstream nose; outer surface (12 mm plate shell assumed).

Pt	Axial a	Ø = 2r
1	464.0	888.9
2	380.0	856.0
3	300.3	801.9
4	226.7	728.1
5	161.1	636.4
6	105.0	529.0
7	59.9	408.6
8	26.9	278.1
9	6.8	140.8
10	0.0	0.0
11	1390.0	0.0
12	1370.0	900.0
13	550.0	900.0

1. Shell of S355J2+N plate 12 mm, formed in gores and rings, full-penetration butt welds (EN ISO 5817 level B), internal ring frames at 400 mm.

2. Watertight: air test at 0.5 bar with soap solution on all seams, then 0.2 bar hold for 1 h without loss.

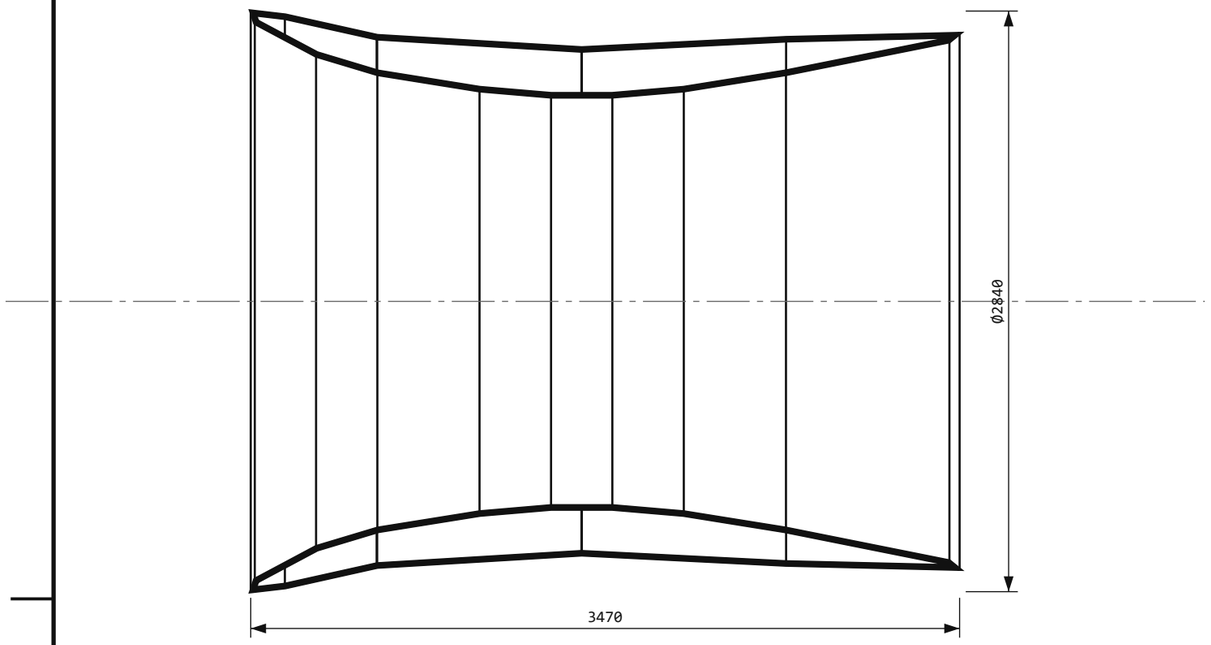
3. Outer surface fair within ±3 mm of the profile over any 1 m chord; coating per general note 4.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

BULB NACELLE

Watertight shell housing the generator

Drawing no. **TL-KAP-006**	Sheet **7 / 13**	Scale (A3) **1:15**	Rev. **A**
Material **S355J2+N plate, EN 10025-2**	Mass **488 kg**	Units **mm**	Date **2026-11-06**
General tol. **ISO 2768-mK**	Welds **ISO 13920-BF**	Projection	Drawn **Turbine Lab**



SECTION

on the axis · scale 1:30

PROFILE COORDINATES

Closed profile of the duct wall: points around the section (a from the inlet).

Pt	Axial a	Ø = 2r
1	620.0	2600.0
2	1620.0	2480.0
3	2620.0	2580.0
4	3470.0	2620.0
5	3420.0	2540.0
6	2620.0	2220.0
7	2120.0	2060.0
8	1770.0	2000.0

Pt	Axial a	Ø = 2r
9	1470.0	2000.0
10	1120.0	2060.0
11	620.0	2220.0
12	320.0	2400.0
13	20.0	2720.0
14	0.0	2840.0
15	170.0	2800.0

1. Fabricate as a full 360° ring in 4 rolled segments of S355J2+N plate 16 mm skin with internal ring and longitudinal stiffeners, welded to EN ISO 3834-2.

2. Inner surface at the runner (throat) machined after welding to Ø1978 +2/0 mm, roundness 2 mm, for the 3 mm tip clearance.

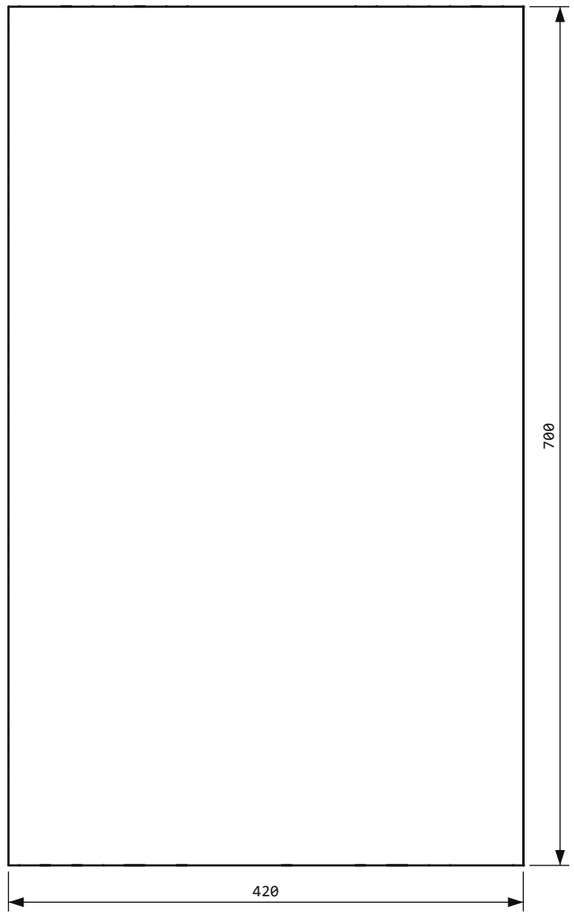
3. Leading and trailing edges rounded as the profile; coating per general note 4; anodes on the outer skin.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

DUCT (SHROUD)

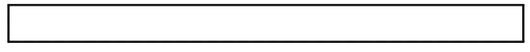
Full 360° ring — the model shows a 270° cutaway

Drawing no. **TL-KAP-007**	Sheet **8 / 13**	Scale (A3) **1:30**	Rev. **A**
Material **S355J2+N plate, EN 10025-2**	Mass **7.65 t**	Units **mm**	Date **2026-11-06**
General tol. **ISO 2768-mK**	Welds **ISO 13920-BF**	Projection	Drawn **Turbine Lab**



VIEW ON THE PLATE

scale 1:5



EDGE VIEW

plate t = 30 · scale 1:5

NOTES

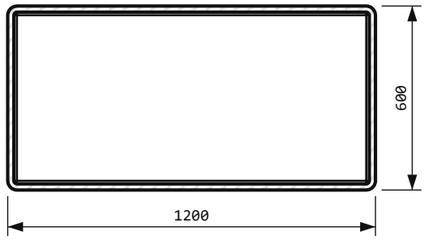
1. Plate S355J2+N, t = 30 mm (from the model), profile cut by plasma or waterjet; edges ground to R 3.
2. 8 off, at 45° spacing about the runner axis; welded both sides to the bulb and the duct with fillet welds a = 8, EN ISO 5817 level B.
3. Flatness 2 mm over the plate; position of each vane ±3 mm, angle ±0.5°, checked against the bulb axis before welding.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

GUIDE VANE

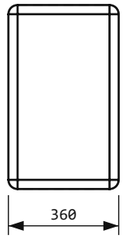
Flat plate vane between bulb and duct, 8 off

Drawing no. TL-KAP-008	Sheet 9 / 13	Scale (A3) 1:5	Rev. A
Material S355J2+N plate, EN 10025-2	Mass 69 kg	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab



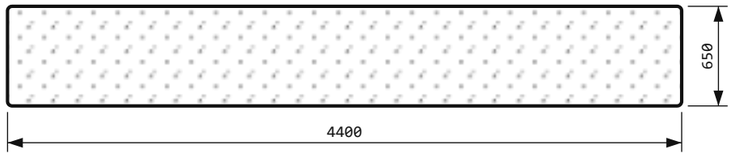
PYLON — SECTION

scale 1:20



PYLON — END

scale 1:20



BASE — SECTION

scale 1:40



BASE — PLAN

scale 1:40

NOTES

1. Pylon: welded box of S355J2+N plate 20 mm, internal diaphragm at mid-height; edges chamfered 30 mm as modelled; top flange bolted to the bulb/duct frame.
2. Base: precast concrete C35/45 (EN 206), exposure class XC4/XF3, reinforcement B500B with 60 mm cover; edges chamfered 30 mm as modelled.
3. Base mass 24.69 t from the model volume (10.29 m³ at 2400 kg/m³). Check against sliding and overturning for the design flow; add ballast or scour protection as required.
4. Cast-in anchor plate for the pylon, position from the model: pylon centre at x = -300, y = 0.
5. Lifting: 4 cast-in lifting anchors, rated for 1.5 × the base mass.

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

SUPPORT PYLON AND GRAVITY BASE

Structure carrying the unit on the river bed

Drawing no.	Sheet	Scale (A3)	Rev.
TL-KAP-009	10 / 13	1:20 / 1:40	A
Material	**Mass**	**Units**	**Date**
items 7 and 8	25.09 t	mm	2026-11-06
General tol.	**Welds**	**Projection**	**Drawn**
ISO 2768-mK	ISO 13920-BF		Turbine Lab

Item 5 — Runner blade (5 off)

Op	Work centre	Operation	Check
10	Pattern shop	3D-printed sand mould from step/blade.step with 4 mm machining allowance on both faces	Mould dims
20	Foundry	Cast CA6NM; one test bar per heat	Chem. analysis
30	Heat treatment	Normalise 1000–1050 °C, temper 590–620 °C	Hardness, tensile
40	NDT	Visual, PT (ISO 3452), UT of root (ASTM A609); weld-repair to procedure and re-heat-treat	NDT report
50	5-axis milling	Rough then finish both faces and edges to the CAD model; leave root datum	CMM vs CAD ±1.0
60	Bench	Blend edges, polish to Ra ≤ 3.2 µm; mark blade number	Templates G–K
70	Machine shop	Machine the root face and bolt holes to the trunnion	Datum check
80	Inspection	Final PT; weigh each blade; blades within 1 % of each other	Mass record

Item 6 — Blade trunnion (5 off)

Op	Work centre	Operation	Check
10	Forge	Open-die forge 42CrMo4 disc	Forging cert.
20	Heat treatment	Quench and temper to 900–1100 MPa	Hardness
30	NDT	UT to EN 10228-3 class 3	UT report
40	CNC lathe	Turn all over to drawing TL-KAP-004; seat to h7	Seat Ø, flatness
50	Machining centre	Drill and tap bolt pattern, pin bore	Positions

Item 4 — Runner hub (1 off)

Op	Work centre	Operation	Check
10	Pattern shop	Pattern and cores from the hub profile (TL-KAP-005), 60 mm wall	Pattern dims
20	Foundry	Cast G20Mn5; heat treat (QT)	Cert., hardness
30	NDT	UT of seat zones, MT after machining	NDT report
40	Vertical lathe	Turn outer profile and the shaft flange face	Profile ±0.5
50	Horizontal boring	Bore 5 trunnion seats H7 at 72°, faces square	True position Ø0.1
60	Assembly	Fit trunnions and blades; set blade angle ±0.1°; balance G6.3 (ISO 21940-11)	Balance report

TEACHING MODEL — materials, tolerances and wall thicknesses are assumptions. Not for construction.

MANUFACTURING PROCESS PLAN — RUNNER

Operation sequence for items 4, 5 and 6

Drawing no. TL-KAP-010	Sheet 11 / 13	Scale (A3) —	Rev. A
Material —	Mass —	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab

Item 1 — Duct (1 off)

Op	Work centre	Operation	Check
10	Cutting	Nest and cut plate S355 16 mm, 4 segments plus stiffeners	Plate certs 3.1
20	Rolling	Roll each segment to the duct profile (TL-KAP-007)	Templates ±3
30	Welding	Assemble segments on a jig, weld seams and stiffeners (WPS)	VT, UT 10 %
40	Machining	Machine the throat Ø to +2/0, roundness 2 mm	Ø at 8 positions
50	Coating	Blast Sa 2½, epoxy system Im1, fit anodes	DFT readings

Item 2 — Bulb nacelle (1 off)

Op	Work centre	Operation	Check
10	Forming	Press-form 12 mm gores for the nose and tail; roll the cylindrical ring	Templates
20	Welding	Weld gores and rings with internal frames, full penetration	VT, RT/UT 100 % seams
30	Testing	Air test 0.5 bar, hold 0.2 bar for 1 h	Leak test record
40	Coating	As item 1	DFT readings

Item 3 — Guide vane (8 off)

Op	Work centre	Operation	Check
10	Cutting	Waterjet or plasma from plate, t as drawing TL-KAP-008	Outline ±1
20	Bench	Grind edges to R 3	VT

Item 8 — Support pylon (1 off)

Op	Work centre	Operation	Check
10	Cutting	Cut 20 mm plates and diaphragm	Plate certs
20	Welding	Weld box, chamfer edges, weld top flange	VT, MT of corners

Item 7 — Gravity base (1 off)

Op	Work centre	Operation	Check
10	Precast yard	Formwork with chamfer strips; place rebar cage, anchor plate and lifting anchors	Cover 60 mm
20	Casting	Pour C35/45, vibrate, cure 7 days wet	Cubes 7/28 d
30	Inspection	Dimensions, anchor plate position, surface	Survey ±10

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MANUFACTURING PROCESS PLAN — STRUCTURE

Operation sequence for items 1, 2, 3, 7 and 8

Drawing no. TL-KAP-011	Sheet 12 / 13	Scale (A3) —	Rev. A
Material —	Mass —	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab

ASSEMBLY SEQUENCE

1. Runner: fit the five trunnions to the hub seats, mount the blades in numbered order, set all blade angles, torque the bolting; balance the runner (G6.3).
2. Bulb: install the generator and the shaft seal inside the bulb (by the generator supplier); couple the runner to the shaft; check run-out at the hub ≤ 0.2 mm.
3. Guide vanes: set the bulb in the duct on a jig with the axes aligned within 2 mm; position the 8 vanes at 45° and weld them to bulb and duct.
4. Check the tip clearance of 3 ± 1 mm at 8 positions by turning the runner by hand; record.
5. Pylon: weld or bolt the pylon to the duct frame; coat damaged areas.
6. Base: place the gravity base on the prepared river bed, level within 1:200; lower the unit and bolt the pylon to the anchor plate.
7. Connect the export cable; commission: rotation by hand, insulation test, first run at low flow, vibration and temperature monitoring.

INSPECTION AND TEST PLAN

Activity	Standard / criterion	Record	H/W/R
Material certificates (all steel, castings)	EN 10204 type 3.1	Certificates	R
Blade and hub castings: chemistry, mechanical	ASTM A743 CA6NM; EN 10293	Test reports	R
Blade NDT (PT, UT root)	ISO 3452; ASTM A609	NDT reports	W
Blade geometry vs CAD	±1.0 mm profile, templates G–K	CMM report	H
Hub seat bores	H7, true position Ø0.1	Inspection report	W
Welding procedures and welders	EN ISO 15614-1; EN ISO 9606-1	WPQR, certificates	R
Weld inspection (bulb seams 100 %, duct 10 %)	EN ISO 5817 level B	NDT reports	W
Bulb leak test	0.5 bar soap test; 0.2 bar 1 h hold	Test record	H
Runner balancing	ISO 21940-11 grade G6.3	Balance report	H
Tip clearance after assembly	3 ± 1 mm at 8 positions	Measurement sheet	H
Coating	ISO 12944 Im1; dry film thickness	DFT readings	W
Base concrete	EN 206 C35/45; cubes 7/28 days	Lab reports	R

H = hold point (work stops until accepted) · W = witness point · R = document review

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ASSEMBLY SEQUENCE, INSPECTION AND TEST PLAN

From parts to an installed unit

Drawing no. TL-KAP-012	Sheet 13 / 13	Scale (A3) –	Rev. A
Material –	Mass –	Units mm	Date 2026-11-06
General tol. ISO 2768-mK	Welds ISO 13920-BF	Projection	Drawn Turbine Lab